# Times Series Regression and Spectral Analysis on Nitrate Concentrations in Long Island, New York


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## Abstract

Long Island, New York encompasses an area of 1,401 mi$^2$ and contains four counties. Of the four counties Nassau and Suffolk counties make up the largest portion by area. An increase in nitrate concentration can be attributed to certain variables that affect the long-term correlation to the changing nitrate levels .Once the data were obtained, variables were inserted into the Konstanz Information Miner (KNIME), an open source data analytics and data mining software, reporting and integration platform. Spectral analysis was performed to determine periodicities of nitrate concentration of different locations throughout Long Island. All data used was in daily values, and missing data were replaced with the previous daily value. Due to the location of the groundwater divide, Orient Harbor is located at a strategic location where water runoff disperses to the north, while Hog Island Channel region sits furthest south of the groundwater divide and can accumulate a nitrate concentration at a shorter time due to a larger exposed land area. Peconic River sits at a higher potentiometric value than the other two locations, this may cause nitrate to not have adequate time to build up significant nitrate levels due to groundwater intensive moving while in low potentiometric values higher concentrations of nitrate may encouraged. We think, that due to the reduction in watershed exposed surface area from western to eastern Long Island this may have had a noticeable effect on the periodicity of the nitrate variable at each USGS monitoring water station.


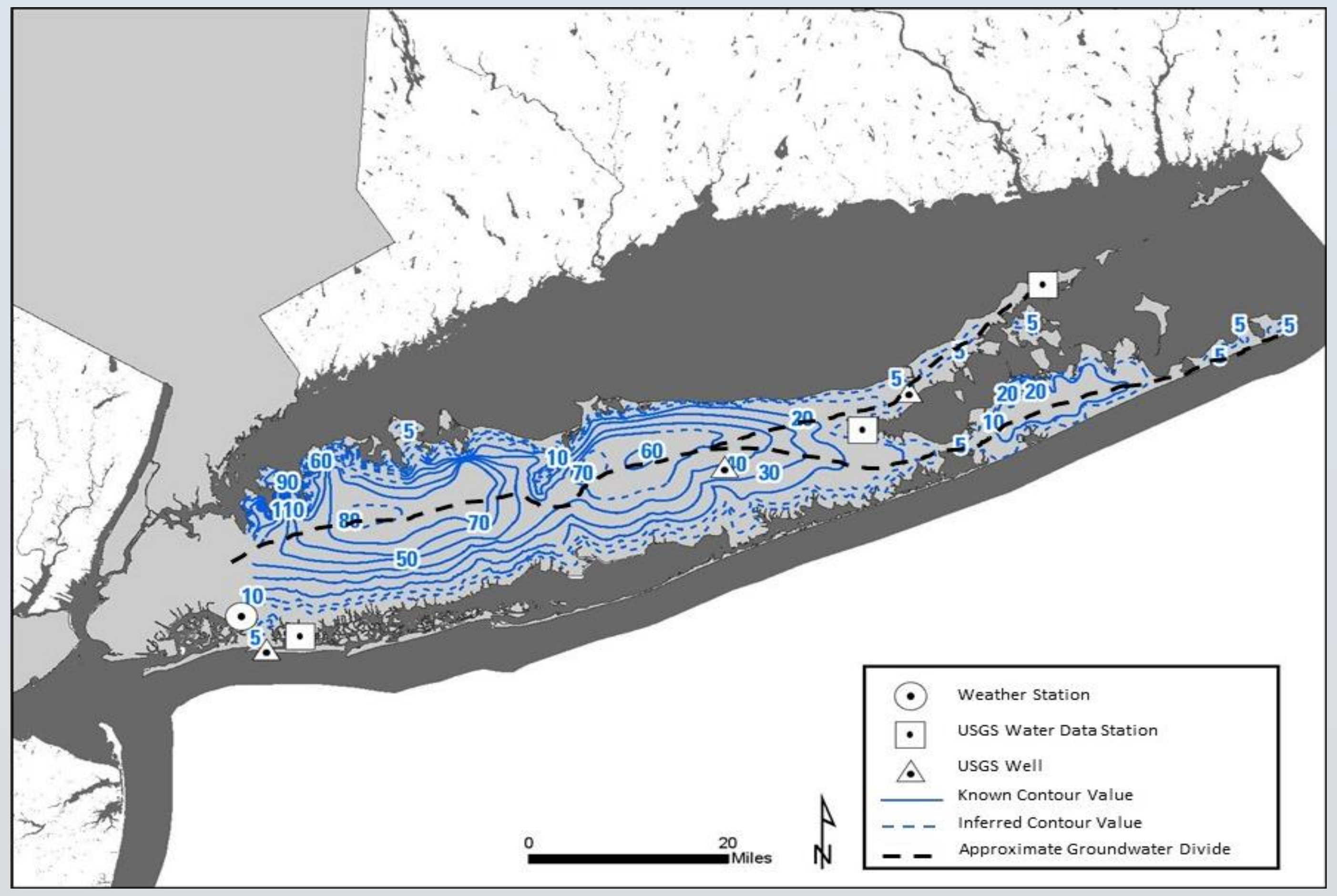


**Figure 1:** Potentiometric contour lines for Nassau and Suffolk County. Long Island's groundwater divide runs west to east and is approximated on this map in dashed black lines. Map contour lines obtained from the USGS [1].

## Introduction

The geologic construct of Long Island has established a series of three main large aquifers to cover the region: the Loyd, Magothy and Upper Glacial aquifers. The extensive coverage and varying depth of these aquifers allow for Long Island residents to use these underground reservoirs to support the growing need for more potable water. Nitrate levels studies on the surface or in the water has been an increasing issue in Long Island and is of major concern to the health of the public due to the use of aquifers as a source of public drinking water [2-6]. An increase in nitrate concentration can be attributed to certain variables that affect the long-term correlation to the changing nitrate levels. It is important to understand what atmospheric and hydrogeological variables have the most effect on the concentration of nitrate near or within the surface soil [7]. Examining Long Island from the east end to west end data was obtained from three United States Geological Survey (USGS) water data stations and three USGS groundwater wells that data were obtained from [8] along with one atmospheric data station from the National Oceanic and Atmospheric Administration [9] .

## Methods

- Data mining and cleaning of atmospheric and hydrogeological variables such as precipitation, depth to groundwater, water temperature, and nitrate concentration. Data was collected for a period of seven years to maintain a common time span for analysis.
- Decomposition of time series and separation of data to create the least interference
- Spectral Analysis to determine periodicity (Fig. 2)
- Raw data decomposed into the long-term component with a periodicity of 177 days selected [10]
- Multiple linear regression was performed

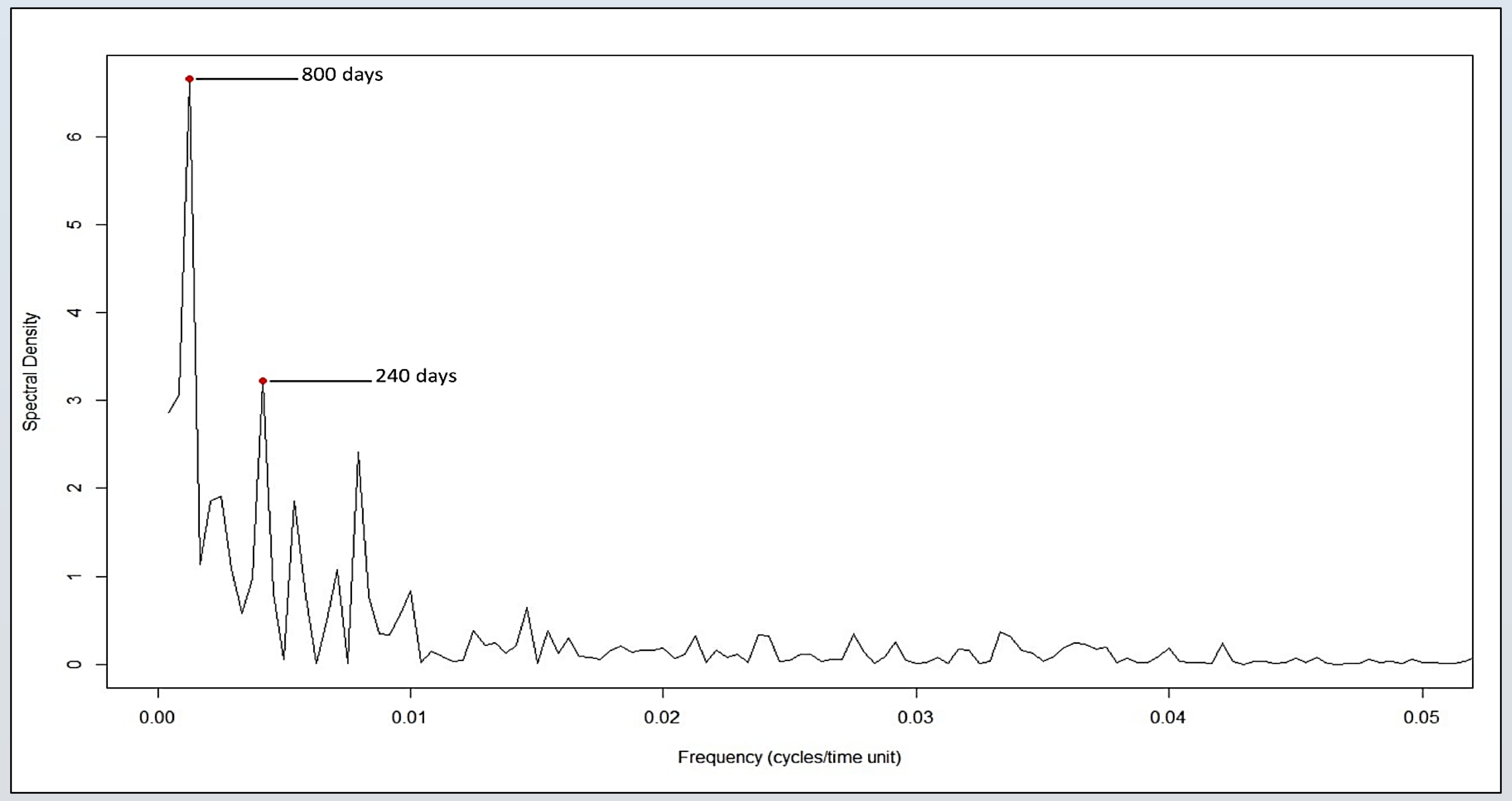


**Figure 2:** Periodograms showing the major cycles of groundwater for Hog Island region. Periodicities (in days) are showing with red bullet on each corresponding spectral density peak.

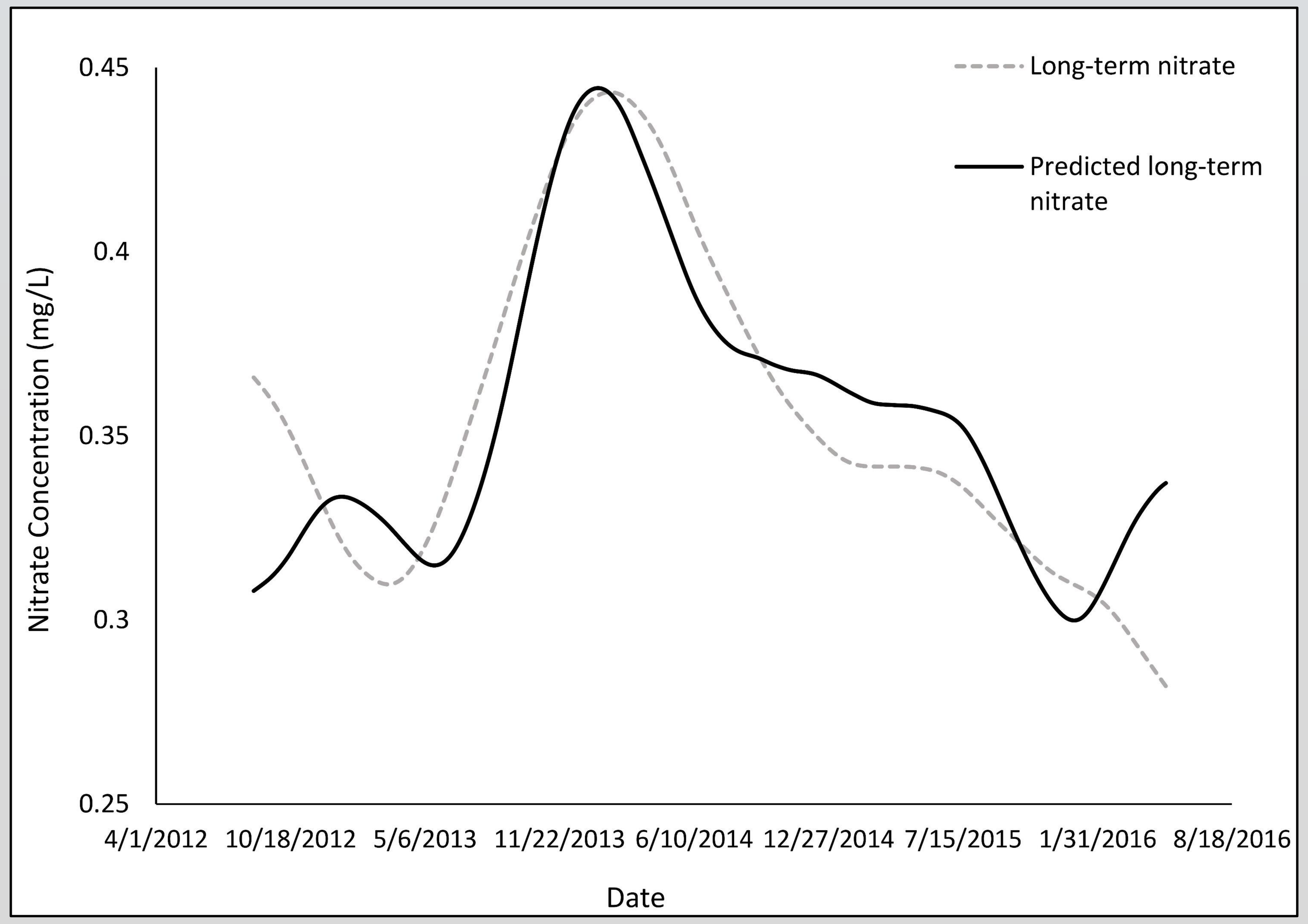


**Figure 3:** The raw nitrate concentration data overlain with the predicted long-term nitrate concentration at Hog Island Channel from April of 2012 to August of 2016. With a long-term correlation value of 0.81 our predicted long-term model describes very well the nitrate concentration based on hydrogeological and atmospheric variations.

## Results

**Table 1:** Periodicity, frequency, and spectral density are presented for the groundwater stations located at Hog Island Channel, NY, Peconic River, NY and Orient Harbor, NY.

| Hog Island | | |
|---|---|---|
| Frequency (cycles/time unit) | Spectral | Period (days) |
| 0.00156 | 2.86 | 640 |
| 0.00625 | 2.85 | 160 |
| **Peconic River** | | |
| Frequency (cycles/time unit) | Spectral | Period (days) |
| 0.00125 | 6.66 | 800 |
| 0.00417 | 3.23 | 240 |
| **Orient Harbor** | | |
| Frequency (cycles/time unit) | Spectral | Period (days) |
| 0.000417 | 3.17 | 2400 |
| 0.00167 | 3.12 | 600 |

## Discussion

- Nitrate concentration periodicity increased as data was examined from data locations further east on Long Island.
- Data limitation to a period of seven years did not allow for exploration of large nitrate cycles, such as a periodicity of 2,400 found at eastern Long Island (Table 1).
- Hog Island has highest $R^2$ due to shortest nitrate periodicity (Fig. 3).
- Large periodicities occur when station distance is shorter from the groundwater. A shorter distance from divide yields less land coverage and less time for nitrate accumulation (Fig. 1).

## Conclusion

Based on a linear regression and spectral analysis performed, a high correlation value of 81% obtained at Hog Island Channel in western Long Island between the nitrate concentration and other atmospheric and hydrogeological variables. As the linear regression model was applied in the other two stations located at the eastern Long Island there was an observable drop in the correlation between nitrate concentration and variables such as precipitation, pH, salinity, water temperature, tidal height and the depth to groundwater. This decrease in correlation can be attributed to a decrease in surface area in eastern Long Island along with each stations location relative to the groundwater divide running west to east across Long Island. We think, that due to the reduction in watershed exposed surface area from western to eastern Long Island this may have had a noticeable effect on the periodicity of the nitrate variable at each USGS monitoring.

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